

Technical Document #204: Data Engineering - Comprehensive Overview

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Data lineage tracks the origin and movement of data through systems. It is essential for debugging, compliance, and impact analysis.

Partitioning divides data into smaller, more manageable pieces. It improves query performance by reducing the amount of data scanned.

Data governance establishes policies and procedures for managing data assets. It covers data privacy, security, and compliance.

Data warehouses store structured data optimized for analytical queries. They use columnar storage and support complex aggregations.

Schema evolution handles changes in data structure over time. Schema registries manage schema versions and ensure backward compatibility.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Key Takeaways:

- Data pipelines automate the movement and transformation of data between systems.
- ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a data warehouse.
- Stream processing handles continuous data streams in real-time.